

Spring 2021 Real Analysis II — Homework 1

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1. Let (X, d) be a metric space. Show that a Cauchy sequence $\{x_k\}$ in X is convergent iff $\{x_k\}$ has a convergent subsequence.
2. Let (X, d) be a complete metric space and $\{x_k\}$ a given sequence. Suppose for any $\epsilon > 0$, there exists a Cauchy sequence $\{y_k\}$ such that $d(x_k, y_k) < \epsilon$ for all $k \in \mathbb{N}$. Show that $\{x_k\}$ is Cauchy.
3. Let (X, d) be a metric space and M a sequentially compact subset of X . Suppose the mapping $f : X \rightarrow M$ satisfies $d(f(x_1), f(x_2)) < d(x_1, x_2)$ for any distinct $x_1, x_2 \in X$. Show that f has a unique fixed point in X .
4. Convergence of Newton's method. Suppose $f \in C^2([a, b])$ and $\hat{x} \in (a, b)$ such that $f(\hat{x}) = 0$ and $f'(\hat{x}) \neq 0$. Show that there exists an open neighborhood $U_{\hat{x}}$ of $\hat{x}$, such that for any $x_0 \in U_{\hat{x}}$, the sequence $\{x_k\}$ generated by Newton's method from x_0 as follows,

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}, \quad k = 1, 2, \dots,$$

is convergent, and $\lim_{k \rightarrow \infty} x_k = \hat{x}$.

5. Let $X = \{f : [0, 1] \rightarrow \mathbb{R} : f(0) = 0, \|f\|_{\text{TV}} < \infty\}$ where

$$\|f\|_{\text{TV}} := \sup_{\Delta} \sum_{i=1}^n |f(x_i) - f(x_{i-1})|,$$

and $\Delta : 0 = x_0 < x_1 < \dots < x_n = 1$ is a partition of $[0, 1]$. Namely, X is the collection of functions who take value 0 at $x = 0$ and has finite total variation. Define $d(f, g) = \|f - g\|_{\text{TV}}$. Show that (X, d) is a complete metric space.

6. Let (X, d) be a metric space, and E and F two compact subsets of X . Show that there exist $x \in E$ and $y \in F$ such that

$$d(x, y) = d(E, F) := \inf_{x' \in E, y' \in F} d(x', y').$$

7. Consider the integral equation

$$x(t) - \lambda \int_0^1 e^{t-s} x(s) \, ds = y(t)$$

where $y \in C([0, 1])$ is given, $\lambda \in \mathbb{R}$ is a constant such that $|\lambda| < 1$. Show that the integral equation has a unique solution $x \in C([0, 1])$. [Hint: multiply e^{-t} on both sides, and show that a unique fixed point of form $e^{-t}x(t)$ exists.]

8. Let M be a bounded subset of $C([a, b])$. Show that the set

$$\left\{ F(x) = \int_a^x f(t) \, dt : f \in M \right\}$$

is sequentially compact. [Hint: use Arzela-Ascoli theorem.]

9. Let $E = \{\sin kt : k \in \mathbb{N}\}$. Show that E is not sequentially compact in $C([0, \pi])$.
10. Let $X := \{x = (x_1, x_2, \dots) : x_i \in \mathbb{R}, \exists K \in \mathbb{N}, \text{ s.t. } x_k = 0, \forall k \geq K\}$, i.e., X is the collection of sequences that have finitely many nonzero terms. Define

$$d(x, y) = \sup_{k \in \mathbb{N}} |x_k - y_k|.$$

Show that $d(x, y)$ is a metric on X , but X is not complete under d . [Hint: consider the sequences of form $(1, \frac{1}{2}, \dots, \frac{1}{k}, 0, \dots)$ at different truncations.]